In [1]:
```
# ============================================================
# Notebook setup
# ============================================================

%load_ext autoreload
%autoreload 2

# Control figure size
figsize=(14, 3.5)
figsize_narrow=(14, 2.5)

use_cached_weights = True

from util import util

import warnings
warnings.filterwarnings("ignore", ".*Consider increasing the value of the `r
warnings.filterwarnings("ignore", ".*GPU available but not used.*")
warnings.filterwarnings("ignore", ".*Previous log files in this directory wi
warnings.filterwarnings("ignore", ".*Checkpoint directory.*exists and is not
```

Loss vs Cost

Prediction and Optimization in the Wild

Many real world decision problems rely on *estimated parameters*

E.g. travel times, demands, item weights/costs...

- Sometimes, these are fixed and can be computed via statistics
- Sometimes, instead, we have access to *a bit more information*

Prediction and Optimization in the Wild

Take *traffic-dependent travel times* as an example

If we know the *time of the day* we can probably estimate them better

Let's see how these problems are often addressed

Predict...

First, we train *an estimator for the problem parameters*:

$$\theta^* = \text{argmin}_\theta \{\mathbb{E}_{(x,y)\sim P(X,Y)}[L(y, \hat{y})] \mid \hat{y} = h(x, \theta)\}$$

- L is the loss function (usually negative log likelihood)
- h is the estimator, with parameter vector θ
- $P(X, Y)$ is the data distribution
- ...Which will typically be approximated via a sample (training set)

In our example:

- x would be the time of the day
- y would be a vector of travel times

...Then Optimize

Then, we *solve an optimization problem* with the estimated parameters

We will start by considering the formulation from this seminal paper

$$z^*(\hat{y}) = \text{argmin}_z \left\{\hat{y}^T z \mid z \in F\right\}$$

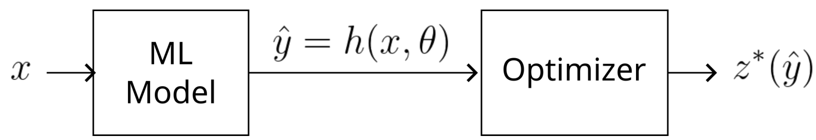
- z is the vector of variables of the optimization problem
- The cost function is linear
- F is the feasible space
- In general, both y and F may depend on the estimated parameters

In our example

- z may represent routing decisions
- the cost might be (e.g.) the total travel time
- F may encode a deadline constraint

Inference

This setup involves using the estimator and the optimizer *in sequence*



At inference time we evaluate:

$$z^*(h(x; \theta))$$

- We observe x
- We evaluate our estimator $h(x; \theta)$ to obtain $\hat{y}$
- We solve the problem to obtain $z^*(\hat{y})$

Prediction Focused Learning

This two-stage approach used to have no name at all

These days, it is referred to as:

- Predict, then Optimize
- ...Or *Prediction Focused Learning*

PFL has several favorable properties:

- It's *easy to implement*
- ...It has *good scalability*
- ...And it's asymptotically correct (perfect predictions result in minimum cost)

Application fields include logistics, planning, finance, etc.

However, the method has also a significant flaw

One that was only recently emphasized

A Toy Problem

Let's see this in action on a toy problem

Consider this two-variable optimization problem:

$$\mathrm{argmin}_z \{y_0 z_0 + y_1 z_1 \mid z_0 + z_1 = 1, z \in \{0,1\}^2\}$$

Let's assume that the true relation between x (a scalar) and y is:

$$y_0 = 2.5x^2 \tag{1}$$
$$y_1 = 0.3 + 0.8x \tag{2}$$

...But that we can only learn the following ML model with a scalar weight θ:

$$\hat{y}_0 = \theta^2 x \tag{3}$$
$$\hat{y}_1 = 0.5\theta \tag{4}$$

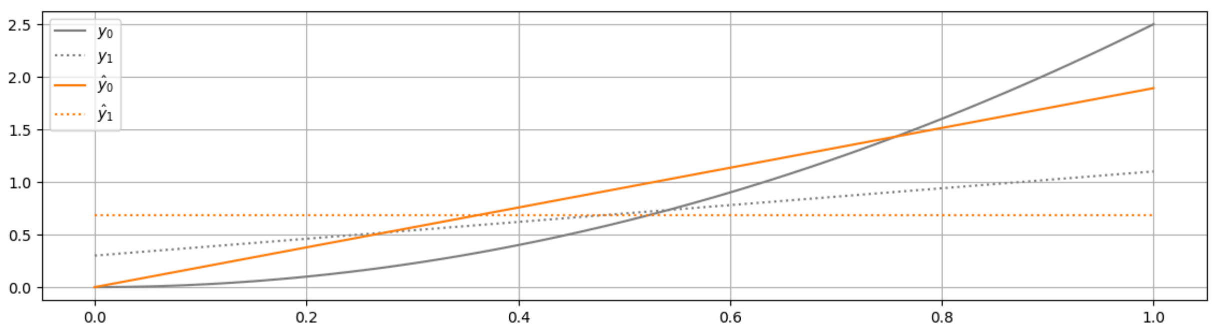
Our model *cannot* represent the true relation exactly

Spotting Trouble

This is what we get from supervised learning with uniformly distributed data:

In [2]:
```
util.draw(w=None, figsize=figsize, model=1)
```

```
Optimized theta: 1.375
```



- The crossing point of the grey lines is where we should pick item 0 instead of 1
- The orange lines (trained model) miss it *by a wide margin*

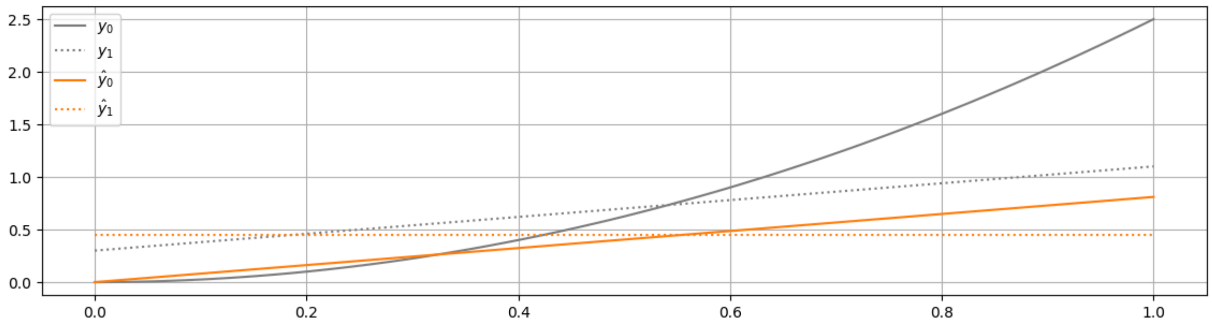
Hence, if we optimize based on our best predictions, we make mistakes!

...But why is this happening?

Misaligned Objectives

We trained for *maximum accuracy* regardless of the decision cost!

In [3]:
```
util.draw(w=0.9, figsize=figsize, model=1)
```



- However, if we focus on choosing θ to *match the crossing point*
- ...The same model can lead the optimizer consistently to the correct choice

We just need to train for *minimanl decision cost*, which is key idea in *Decision Focused Learning*

The y parameters *cannot be measured*

...But they depend on *some observable x*

- And both can be represented as random variables with a joint distribution:

$$X, Y \sim P(X, Y)$$

Getting Started

The y parameters *cannot be measured*

...But they depend on *some observable x*

- And both can be represented as random variables with a joint distribution:

$$X, Y \sim P(X, Y)$$

So, in practice we can estimate y via a ML model

$$\hat{y} = h(x; \theta)$$

...And at inference time we get our decisions by computing:

$$z^*(h(x; \theta))$$

I.e. exactly as in Prediction Focused Learning

The Main DFL Idea

The key difference between PFL and DFL is *the training process*

...Whic in DFL is done by minimizing a *decision cost*, i.e. by solving:

$$\theta^* = \text{argmin}_\theta \{\mathbb{E}_{(x,y)\sim P(X,Y)}[\text{regret}(y, \hat{y})] \mid \hat{y} = h(x, \theta)\}$$

Where in our setting we have:

$$\text{regret}(y, \hat{y}) = y^T z^*(\hat{y}) - y^T z^*(y)$$

- $z^*(\hat{y})$ is the best solution with the *estimated* costs
- $z^*(y)$ is the best solution with the *true* costs

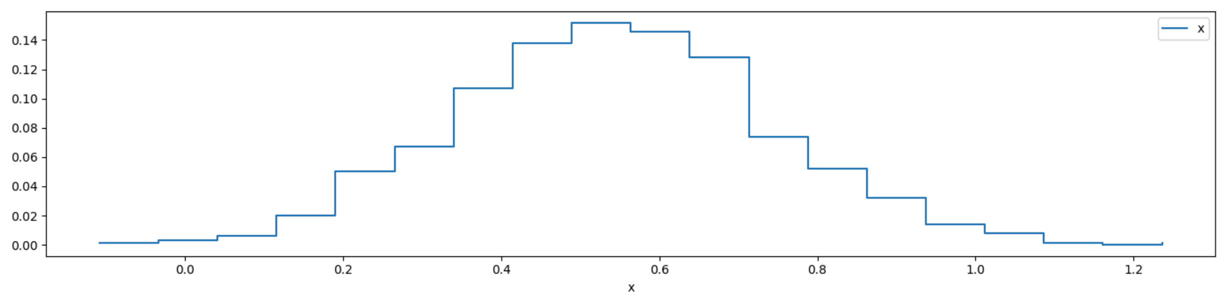
Intuitively, we want to *loose as little as possible* w.r.t. the best we could do

One of the main challenges in DFL is dealing with this loss

Knowing Regret

To see this, let's push our example a little further

In [4]:
```
x = util.normal_sample_(mean=0.54, std=0.2, size=1000)
util.plot_histogram(x, figsize=figsize, label='x');
```

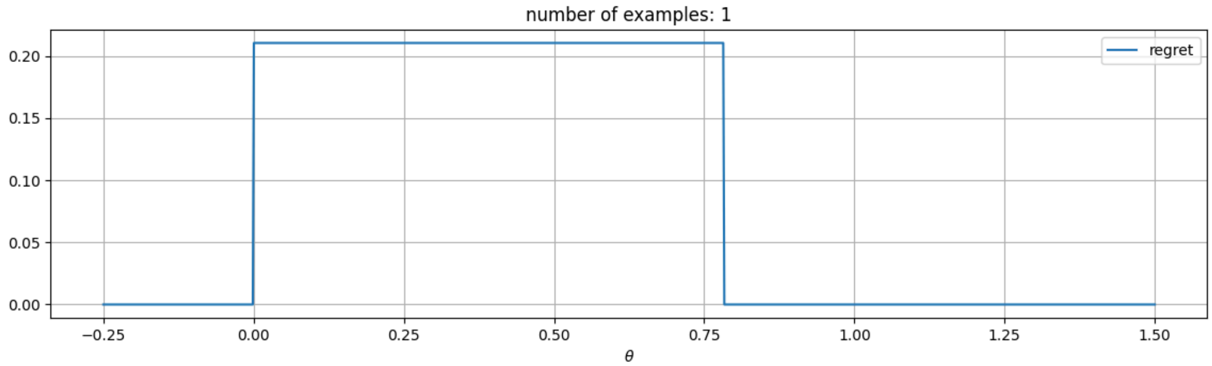


- Say we have access to a normally distributed collection of x values
- ...And to the corresponding true values y

Knowing Regret

This is how the regret looks like for a single example

In [5]:
```
util.draw_loss_landscape(losses=[util.RegretLoss()], model=1, seed=42, batch
```

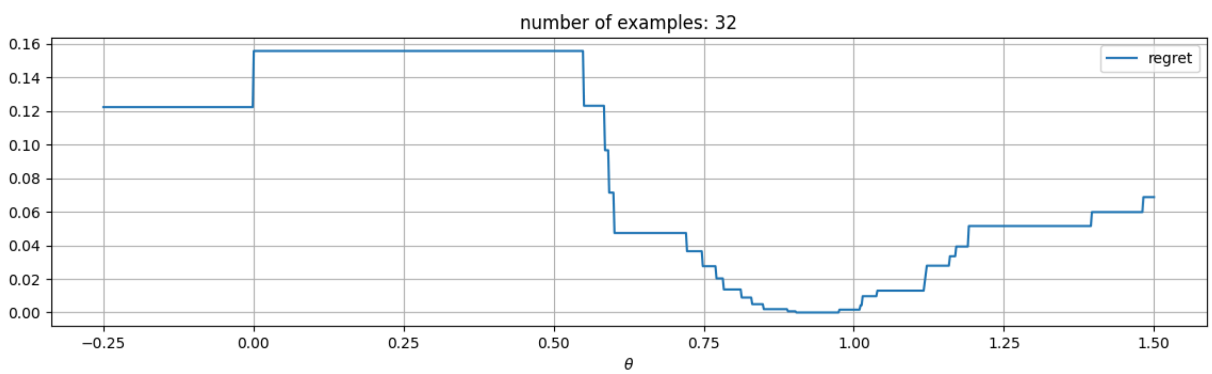


- If $f(x, \theta)$ leads to the correct decision, the regret is 0
- Otherwise we have some non-null value

Knowing Regret

...And this is the same for a larger sample

In [6]:
```
util.draw_loss_landscape(losses=[util.RegretLoss()], model=1, seed=42, batch
```



This function breaks havoc with gradient descent, for *two main reasons*

What's Wrong with Regret (1)

As a first issue, the loss is *not differentiable in general*

Given:

$$\text{regret}(y, \hat{y}) = y^T z^*(\hat{y}) - y^T z^*(y) \qquad \text{with: } \hat{y} = h(x; \theta)$$

The derivative chain:

$$\boxed{\frac{\partial\, \text{regret}}{\partial \theta} = \frac{\partial\, \text{regret}}{\partial z^*} {\color{red}\frac{\partial\, z^*}{\partial \hat{y}}} \frac{\partial\, \hat{y}}{\partial \theta}}$$

...Contains a term that is based on an argmin operator

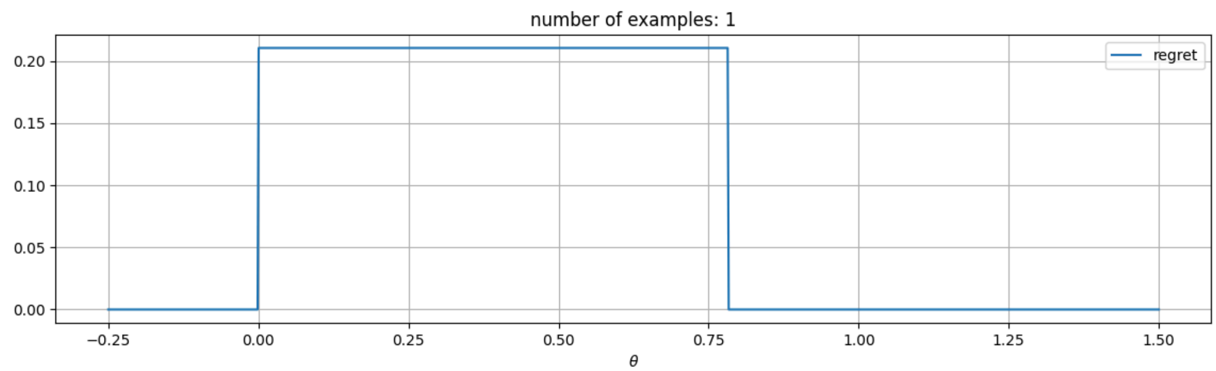
- For this reason, computing the derivative might be tricky
- ...And for some $\hat{y}$ values a derivative might not be defined at all

What's Wrong with Regret (2)

Second, when the derivative exists, it might be *useless*

E.g. for combinatorial and linear problems, regret will be *piecewise constant*

In [7]:
```
util.draw_loss_landscape(losses=[util.RegretLoss()], model=1, seed=42, batch
```



When the derivative is defined, *its value is 0*

Self-Contrastive Loss

These issues have been addressed in multiple ways

Here we'll start with the idea of *changing perspective*

- In particular, any prediction vector $\hat{y}$ defines a cost function:

$$\hat{y}^T z$$

- ...Which will lead the solver toward the optimal solution:

$$z^*(\hat{y})$$

Given an example (x, y), for a *good* prediction vector $\hat{y}$

- The cost of the *true* optimal solution $z^*(y)$
- ...Should not be worse than the cost of the *"estimated"* optimal solution $z^*(\hat{y})$

Self-Contrastive Loss

Hence we can think of using as a *surrogate loss* the difference:

$$\hat{y}^T z^*(y) - \hat{y}^T z^*(\hat{y})$$

It represents "how wrong" the estimated cost function is w.r.t. the true one

- It contains a naturally differentiable term (i.e. $\hat{y}$)
- It is not constant, even when z^* is piecewise constant

Differencing over $\hat{y}$ gives a subgradient

...Which in this case is given by the difference between the solutions

$$\nabla \left(\hat{y}^T z^*(y) - \hat{y}^T z^*(\hat{y})\right) = z^*(y) - z^*(\hat{y})$$

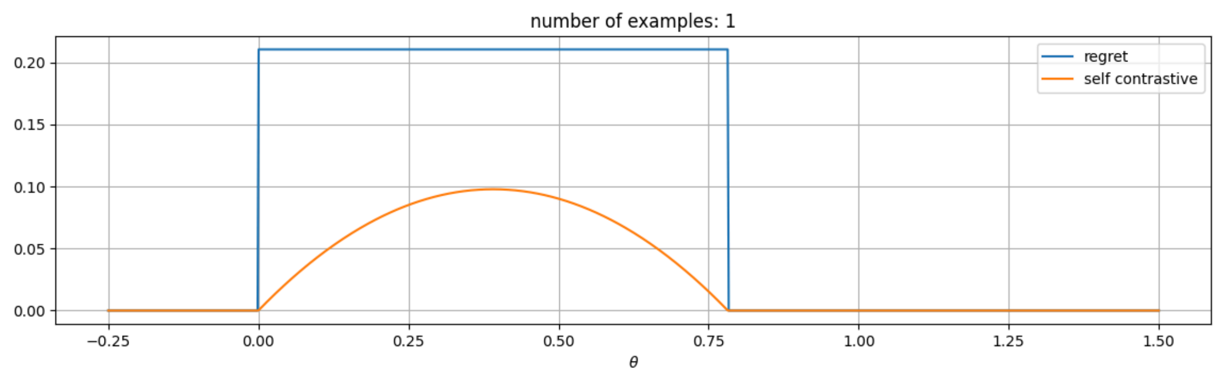
In the DFL literature, this is known as *self-contrastive loss*

Limitations of the Self-Contrastive Loss

However, the self-constrastive loss has some significant limitations

Here's how it looks for one example in our toy problem:

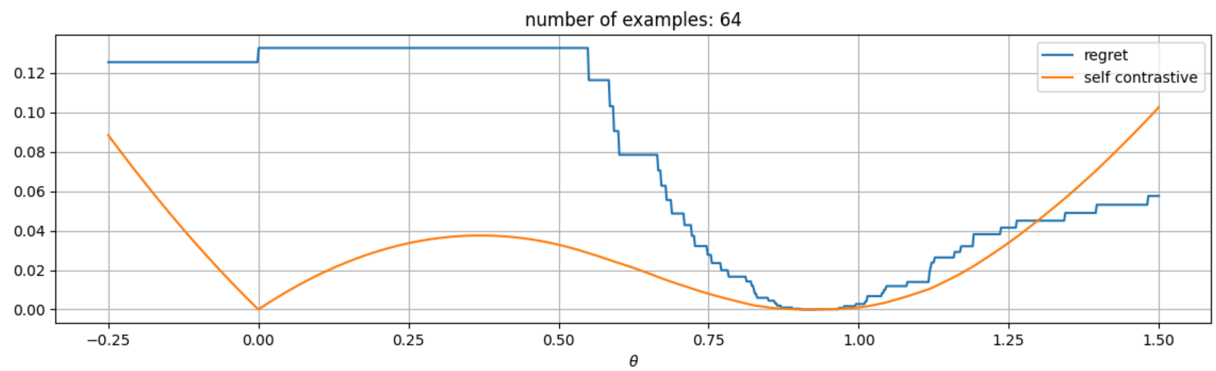
In [8]:
```
util.draw_loss_landscape(losses=[util.RegretLoss(), util.SelfContrastiveLoss
```



Limitations of the Self-Contrastive Loss

Here's the plot for multiple examples

In [9]:
```
util.draw_loss_landscape(losses=[util.RegretLoss(), util.SelfContrastiveLoss
```

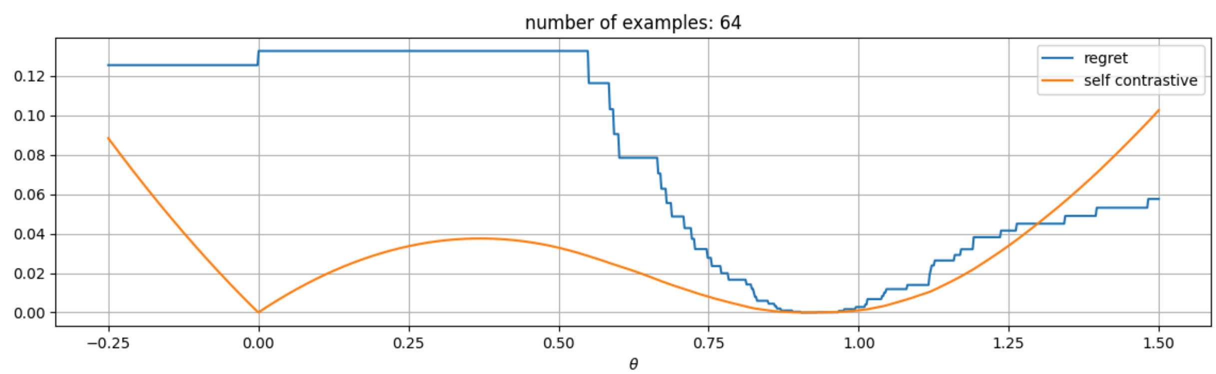


There's a problem here! **Can you see which one?**

Limitations of the Self-Contrastive Loss

There's *a spurious minimum*!

In [10]:
```
util.draw_loss_landscape(losses=[util.RegretLoss(), util.SelfContrastiveLoss
```



At training time, there's a chance of reaching *the wrong minimum*!

SPO+ Loss

A well-known DFL approach can be seen as solution for this issue

I.e. the SPO+ loss from this paper, which can be defined as:

$$\text{spo}^+(y, \hat{y}) = \hat{y}_{spo}^T z^*(y) - \hat{y}_{spo}^T z^*(\hat{y}_{spo}) \quad \text{with:} \quad \hat{y}_{spo} = 2\hat{y} - y$$

- The structure is similar to the self-constrastive loss
- ...But *at training time* we compute it w.r.t. a perturbed prediction vector

At inference time we behave as usual, i.e. we solve:

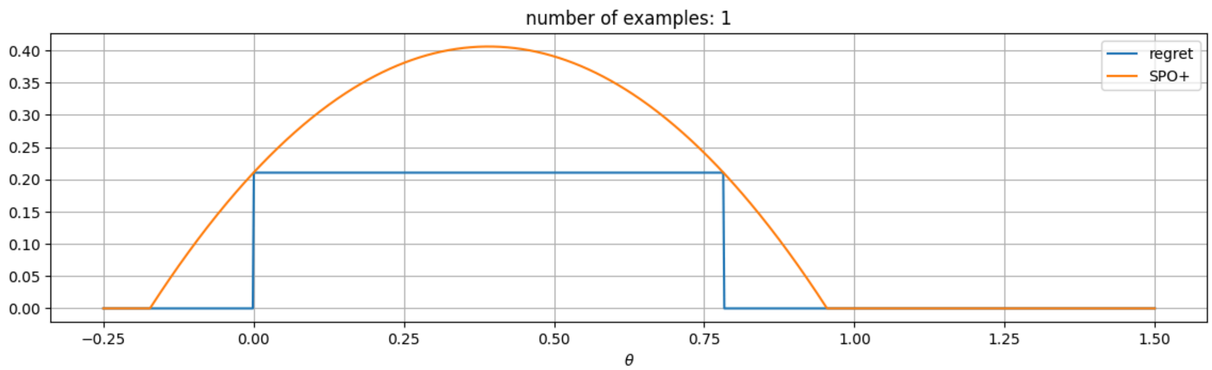
$$z^*(\hat{y}) \quad \text{with: } \hat{y} = h(x; \theta)$$

This will still result in an improvement if training lead us to a good local optimum

SPO+ Loss

This is the SPO+ loss for a single example on our toy problem

In [11]:
```
util.draw_loss_landscape(losses=[util.RegretLoss(), util.SPOPlusLoss()], moc
```

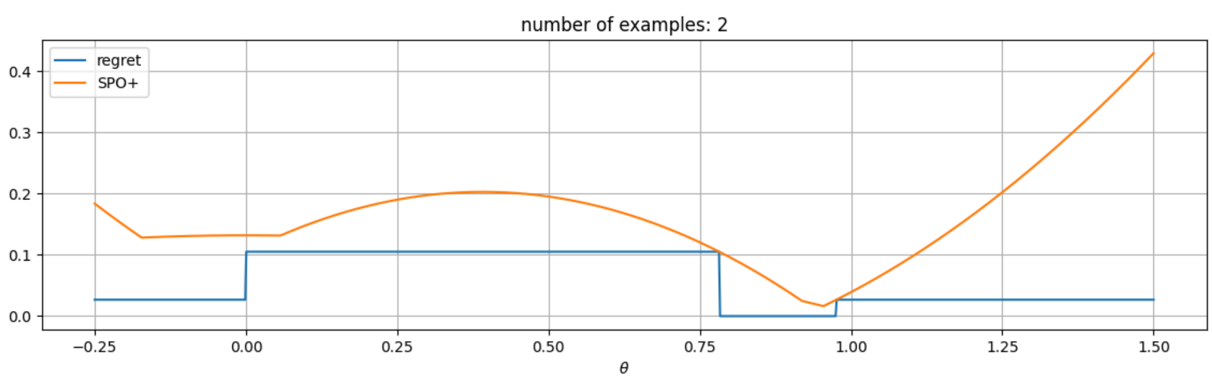


- Like in the self-constrastive case, there are two local minima

SPO+ Loss

This is the SPO+ loss for a *two examples*

In [12]:
```
util.draw_loss_landscape(losses=[util.RegretLoss(), util.SPOPlusLoss()], moc
```

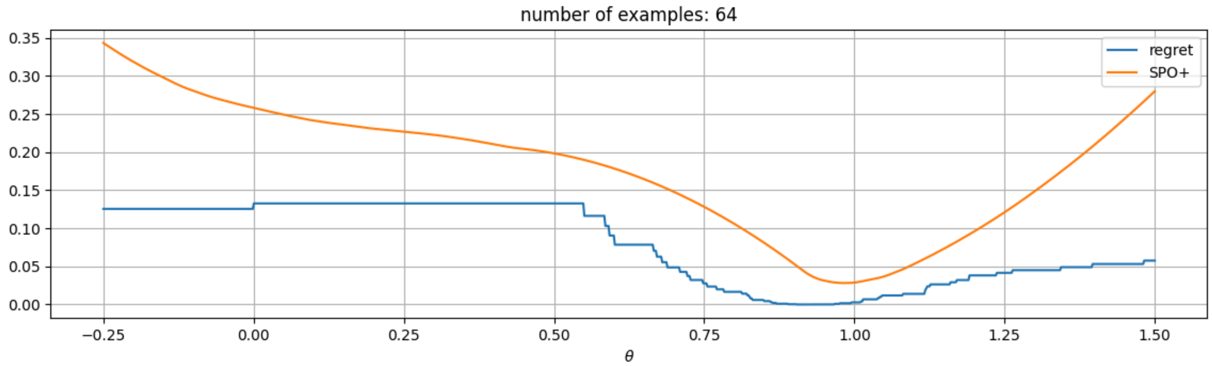


- The "good" local minima for both examples are roughly in the same place
- The "spurious" local minima fall in different position

SPO+ Loss

Over many example, the spurious local minima tend to cancel out

In [13]:
```
util.draw_loss_landscape(losses=[util.RegretLoss(), util.SPOPlusLoss()], moc
```



- This effect is *invaluable* when training with gradient descent

Let's see the approach in action on a single problem

A (Sligthly) More Complex Example

We will consider an optimization problem in this form:

$$z^*(y) = \text{argmin}\{y^T z \mid v^T z \geq r, z \in \{0, 1\}^n\}$$

- We need to decide which of a set of jobs to accept
- Accepting a job ($z_j = 1$) provides immediate value v_j
- The cost y_j of the job is not known
- ...But it can be estimated based on available data

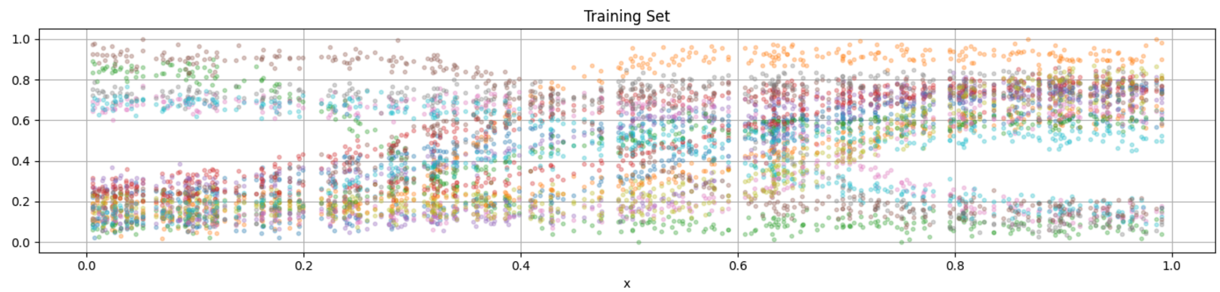
In [14]:
```
nitems, rel_req, seed = 20, 0.5, 42
prb = util.generate_problem(nitems=nitems, rel_req=rel_req, seed=seed)
display(prb)
```

```
ProductionProblem(values=[1.14981605 1.38028572 1.29279758 1.23946339 1.0624
0746 1.06239781
 1.02323344 1.34647046 1.240446   1.28322903 1.0082338  1.38796394
 1.33297706 1.08493564 1.07272999 1.0733618  1.1216969  1.20990257
 1.17277801 1.11649166], requirement=11.830809153591138)
```

A (Sligthly) More Complex Example

Next, we generate some training (and test) data

In [15]:
```
data_tr = util.generate_costs(nsamples=350, nitems=nitems, seed=seed, noise_
data_ts = util.generate_costs(nsamples=150, nitems=nitems, seed=seed, sampli
util.plot_df_cols(data_tr, figsize=figsize, title='Training Set', scatter=Tr
```

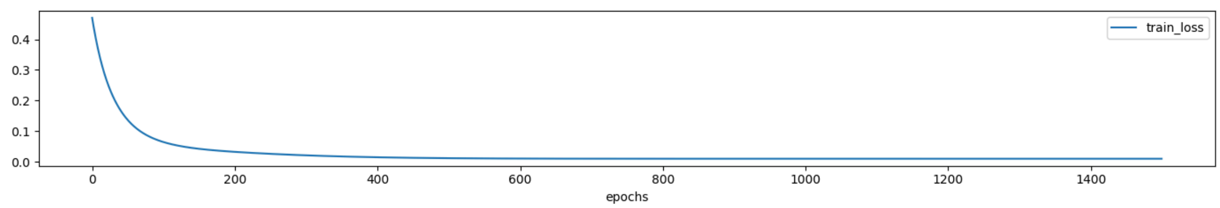


- We assume that costs can be estimated based on an scalar observable x
- The set of least expensive jobs changes considerably with x

Prediction Focused Approach

As a baseline, we'll consider a basic prediction-focused approach

In [16]:
```
pfl = util.TwoPhaseModel(input_size=1, output_size=nitems, hidden_sizes=[], 
pfl, history, metadata = util.train_ml_model(pfl, data_tr.index.values.resha
util.plot_training_history(history, metadata=metadata, figsize=figsize_narro
util.print_ml_metrics(pfl, data_tr.index.values.reshape(-1, 1), data_tr.valu
util.print_ml_metrics(pfl, data_ts.index.values.reshape(-1, 1), data_ts.valu
```



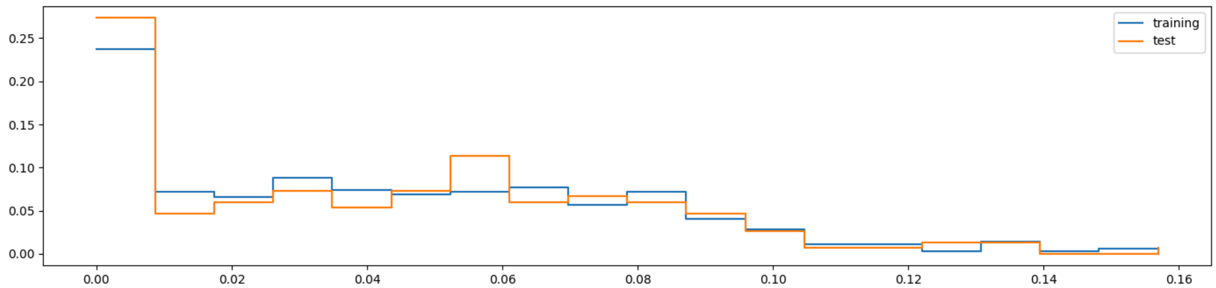

```
training time: 9.5323
R2: 0.829, MAE: 0.081, RMSE: 0.099 (training)
R2: 0.823, MAE: 0.083, RMSE: 0.101 (test)
```

- The ML model is just a linear regressor, but it is decently accurate

Prediction Focused Approach

...But our true evaluation should be in terms of *regret*

In [17]:
```
r_tr = util.compute_regret(prb, pfl, data_tr.index.values.reshape(-1, 1), da
r_ts = util.compute_regret(prb, pfl, data_ts.index.values.reshape(-1, 1), da
util.plot_histogram(r_tr, figsize=figsize, label='training', data2=r_ts, lab
```



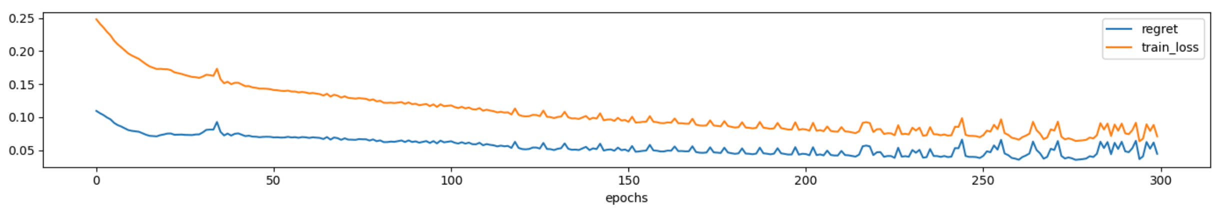

```
Mean: 0.043 (training), 0.043 (test)
```

- In this case, the average *relative* regret is in the 4-5% range

A Decision Focused Learning Approach

Next, we train a DFL approach (warm started with the PFL weights)

In [18]:
```
spo = util.DFLModel(input_size=1, output_size=nitems, hidden_sizes=[], probl
spo, history, metadata = util.train_ml_model(spo, data_tr.index.values.resha
util.plot_training_history(history, metadata=metadata, figsize=figsize_narro
util.print_ml_metrics(spo, data_tr.index.values.reshape(-1, 1), data_tr.valu
util.print_ml_metrics(spo, data_ts.index.values.reshape(-1, 1), data_ts.valu
```



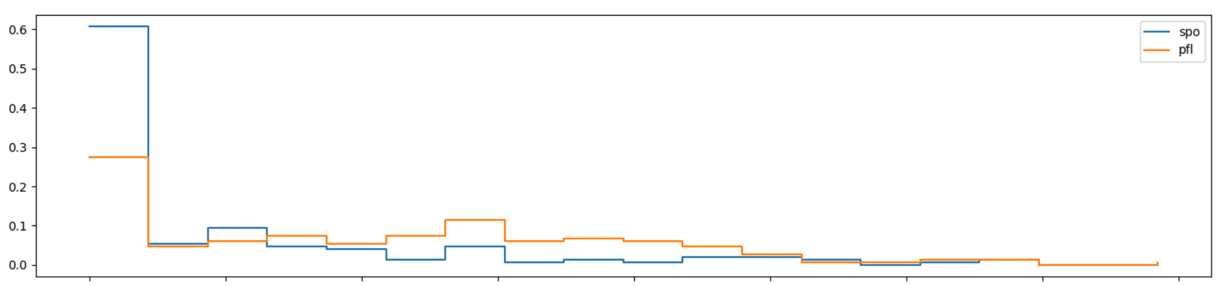

```
training time: 107.4588
R2: 0.705, MAE: 0.099, RMSE: 0.126 (training)
R2: 0.703, MAE: 0.100, RMSE: 0.126 (test)
```

In terms of accuracy, this is considerably worse

Comparing Regrets

But the regret is so much better!

In [19]:
```
r_ts_spo = util.compute_regret(prb, spo, data_ts.index.values.reshape(-1, 1)
util.plot_histogram(r_ts_spo, figsize=figsize, label='spo', data2=r_ts, labe
```



```
Mean: 0.019 (spo), 0.043 (pfl)
```

When this was found some years ago, it was a very surprising result